

THE OHIO Naturalist

PUBLISHED BY
THE BIOLOGICAL CLUB OF THE OHIO STATE UNIVERSITY

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EDITOR-IN-CHIEF—JOHN H. SCHAFFNER, A. M., M. S.

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Volume 1.

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A journal devoted more especially to the natural history of Ohio. The official organ of THE BIOLOGICAL CLUB OF THE OHIO STATE UNIVERSITY.

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The Ohio Naturalist

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No. 5

PERENNIAL TUMBLEWEEDS.

JOHN H. SCHAFFNER.

Tumbleweeds may be classified under three general heads:
Annual tumbleweeds,
Tumble-grasses,
Perennial tumbleweeds.

The annual tumbleweeds are mostly plants with a small root system which shrivels up or rots away soon after the seed has matured. The plants are then easily torn from the ground or broken off and go tumbling away before the wind. In some cases the roots become quite fleshy and brittle. In the tumble-grasses the panicle is generally the only part which is transported, the stems of the panicle being usually very brittle and breaking readily even in those forms which are easily torn up from the roots.

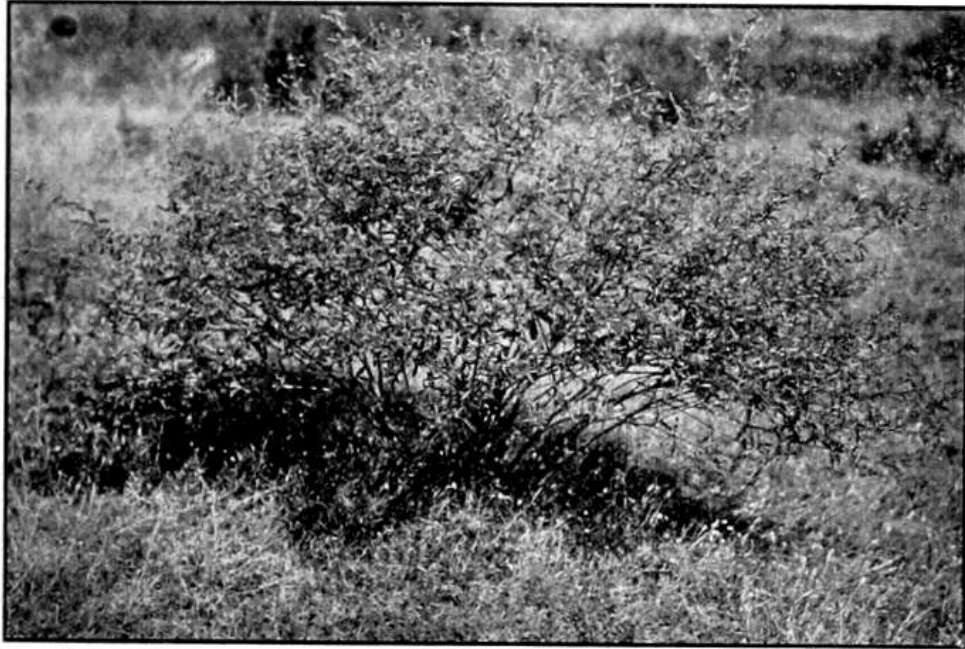


Fig. 1. *Psoralea floribunda*. Plant growing on prairie, Clay Co., Kan.



Fig. 2. Plant of *P. floribunda*, showing a part of the deep taproot.

The perennial tumbleweeds are especially interesting because of the way in which they are separated from the underground parts. Among the perennial forms *Psoralea floribunda* is one of the most typical. It is a long-lived, perennial crown-former with a very deep root which may be several inches in diameter. From the short terminal stem of this root a number of aerial branches are developed annually. These branches take on a more or less globose or balloon-shaped form. At the base of each aerial stem a number of special joints are formed in which transverse cleavage regions are gradually developed, and when the seed is ripe the whole crown breaks off at these joints with remarkable ease. This is a peculiar case of the development of a self-pruning process in the stem for a very special purpose.

Psoralea argophylla also develops perfect joints but fewer shoots usually make up the crown and it is therefore less conspicuous than *P. floribunda*. *Psoralea esculenta* is also a tumbleweed but the writer has not made an examination of the way in which it separates from the thick, tuberous, perennial root.

Psoralea floribunda is very abundant in north-central Kansas where the writer has seen great masses heaped up against hedgerows and wire fences. These plants show a most remarkable responsive adaption to an environment of very definite conditions. They have developed nearly every character possible in harmony with the dry and windy plains of the west and may be regarded as ideal prairie plants.

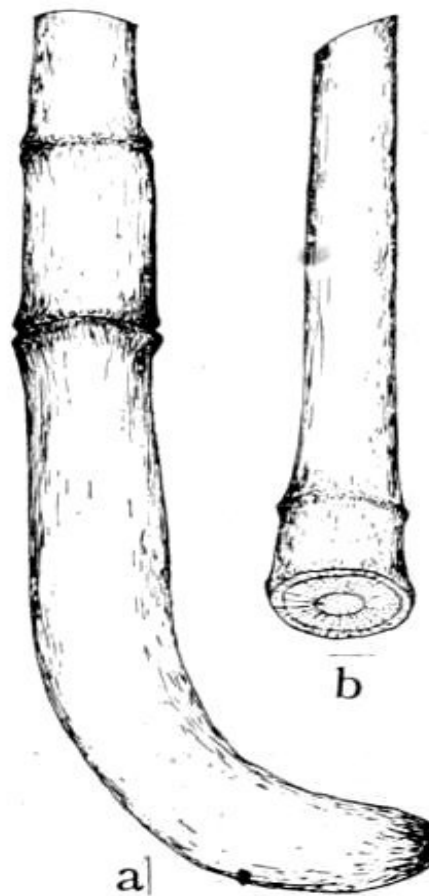


Fig. 3. (a) Base of a stem of *P. floribunda* with two cleavage joints. (b) Base of stem showing cleavage surface.

THE SPROUTING OF COCKLEBUR SEEDS.

E. E. MASTERMAN.

In July, 1896, Dr. E. W. Claypole, then of Buchtel College, Akron, Ohio, asked me how general was the belief that one seed of the cocklebur grew one year and the other the next year or later. Inquiry of about twenty of the older residents resulted in procuring no information touching the same. In 1897, I was told by a German farmer that one seed only grew one year and the other later, never both at the same time. A short time after I noticed the statement of A. D. Selby in Bulletin 83, (page 353) Ohio Experiment Station, as follows: "Prof. Arthur has recently shown that only one of these seeds can be caused to germinate the first year, the other always remaining until the second year." This was a confirmation of the German's claim, yet I determined to investigate for myself.

I carried on the experiment for three years with the following results:

In 1898, I planted 1000 burs; 917 grew two plants to the bur.

In 1899, I planted 1000 burs; 921 grew two plants to the bur.

In 1900, I planted 1000 burs; 913 grew two plants to the bur.

Total three years, 3000 burs; 2751 grew two plants to the bur.

Of the remaining 249 burs some grew one plant, some none; some had one, some had two apparently sound seeds. I regret that no further notice was taken of these seeds. The *only* object was to determine whether the two seeds *could* be made to grow at the same time. An account of the work was sent to Professor Selby, asking whether further experiment was necessary; he replied that he thought not.

Perhaps it should be added that I selected only apparently sound burs; soil was taken from a field near a creek where cockleburs grow abundantly. It was passed through a $\frac{1}{4}$ inch-mesh wire sieve, and carefully searched over with the aid of a glass. This soil was taken to a distant part of the farm; in it the seeds were planted and nature did the rest.

I also made observations as follows: I searched among specimens growing for a mile along a creek, for two plants growing together and not nearer than five inches to any other plant. Of the 1500 specimens examined each year for three years, two plants always grew from one bur.

Why have I obtained such opposite results as compared with Professor Arthur's? Can it be referred to locality, soil, or some other more favorable conditions?

The substance of the above was presented, December 27, 1900, to the Ohio Academy of Science and it provoked a discussion in which Professors Kellerman, Schaffner, Mosely and others participated. Dr. Kellerman thought that the results of Arthur's experiments were perhaps more nearly in accord with what usually takes place in nature. He pointed out the mistake of quoting or saying that Arthur has shown "that only one of the seeds can be caused to germinate the first year." Turning to the printed report of the experiments in question (Proc. 16th, An. Meeting Soc. Prom. Agr. Sci., 1895), I find that, based on many experiments made previous to 1895, he gives the result in round numbers as follows: "Out of every hundred ordinarily well formed cockleburs, seventy will produce one seedling each, and five two seedlings each the first year after maturity; the remaining twenty-five will for various reasons fail to grow. Thirty of the hundred will produce seedlings the second year after maturity, five will produce seedlings the third year after maturity, and two or three seedlings will be produced in subsequent years."

Later experiments by Dr. Arthur seemed to show a lower percentage of cases of the sprouting of both seeds to the bur in one season. In the summary he states: "The germination of both seeds of a bur of *Xanthium* in one season is exceptional."

In view of the above and in accordance with the suggestions of others I purpose continuing my experiments relative to this subject.

The following interesting statement is made by Dr. Arthur, in the report cited, touching the cause of the difference in the action of the two

seeds; he says it “appears to be constitutional; a hereditary character residing in the protoplasm of the embryo.”

New London, Ohio.

PLANT REMAINS FROM THE BAUM VILLAGE SITE.

W. C. MILLS.

During the year 1900 the Ohio Archaeological and Historical Society procured from the ash pits of the Baum Village Site, situated near Bournville, Ross county, Ohio, a number of grains and seeds, which were submitted to Prof. J. H. Schaffner for identification. The following is the list:

Corn, *Zea mays* L.

Great quantities of the eight rowed variety were found. The cobs were usually about one-half inch in diameter. Also a variety with more than eight rows, usually ten rows was found. This variety had a much thicker cob. The grains and cob were in a good state of preservation, having been charred. In several instances the charred remains of a woven fabric were found intermingled with the cobs and grains, showing that the corn had evidently been wrapped in this cloth. In other instances the grains and cobs were found in large pieces of broken pottery and were well preserved. Finding the corn in so many of the pits shows that it largely supplied the food of the camp.

Quantities of charred papaw seeds, *Asimina triloba*, (L.) Dunal, and the wild Hazelnut, *Corylus americana* Walt. were found in a number of pits showing that these were largely used for food.

Quite a quantity of the seeds of the wild red plum, *Prunus americana* Marsh. was also taken from the pits. These were, in a number of instances, associated with papaw seeds and the shells of the chestnut, *Castanea dentata* (Marsh.) Borkh.

Great quantities of the broken shells of the butternut, *Juglans cinerea* L. and the black walnut, *Juglans nigra* L. were discovered. These were usually found associated together, but in several instances they were found separated, the butternuts being more abundant than the walnuts.

Three species of hickory nuts were procured but none of these were in such quantities as the butternut and black walnut. The three species found were as follows: *Hicoria minima* (Marsh.) Britt., *Hicoria ovata* (Mill.) Britt., *Hicoria laciniosa* (Mx.) Britt.

Several specimens of beans, *Phaseolus* (sp) and also a specimen of the grape, *Vitis* (sp.) were found in the material, but it was not possible to tell whether the beans were one of our wild species or cultivated.

SPROUTING FLOWER BUDS OF OPUNTIA.

DR. V. STERKI.

In June of last year I took some *Opuntia* plants home, and also some top joints heavily set with large buds. The former were planted in the garden, the latter set in an Oleander tub. When, after a month, none of the flower buds had opened, it was thought that they were too many, as the joints bearing them were without roots, and most of them were cut off and left lying on the ground, where a part of them later on became partly or entirely covered with soil. In September, I was surprised to find them all green and fresh; most of them had rooted, and a few even sprouted, sending up shoots from half an inch to over an inch high, being perfect little joints. At the present writing (Jan. a. c.) all are alive, and, no doubt, will grow out to plants next summer. They will be watched closely and further report be given.

It might be added that the *Opuntia* calyx-tube, which is later the fruit, has "eyes," that is buds, of the same character as the ordinary buds of the plant, with clusters of bristles; and out of these the young shoots grew, when the bud took root.

Evidently these buds retain more of the nature of the mother plant than is common in flowers. It is unknown to me whether similar observations have been made before. But it would be of interest to make experiments with different plants. Would the receptacles root and sprout if detached after flowering and fertilization have taken place? Would the buds sprout when left in situ on the mother plant, after the flowering parts had been removed, the receptacle only left in place? Will the buds of other genera of *Cactaceae*, and other similar succulent plants behave in the same way, under favorable conditions?

So-called viviparous plants are, as is well known, rather common, e. g. among Gramineae, Cyperaceae, Polygoneae. But there the actual flower parts develop into leaves, from which they had originally been derived, and while yet remaining on the parent plant.

New Philadelphia, Ohio.

NOTE ON THE INVOLUCRAL LEAVES OF SYNDESMON.

F. H. BURGLEHAUS.

Syn-des-mon *thalic-troides* is described in Britton & Brown's Flora as having sessile involucral leaves, which character is contradicted in the plants growing in the vicinity of Toledo. Careful observation during the past season fails to reveal a single instance of sessile involucral leaves, and most of the specimens examined have these leaves borne on petioles from one-fourth to one-half inch in length. Should like to hear from others concerning this feature of one of our most beautiful and dainty spring flowers.

Toledo, Ohio.

COMPETITION IN BOTANY FOR OHIO SCHOOLS.

W. A. KELLERMAN.

Whatever may contribute to a more direct and real study of the plant kingdom on the part of the pupils can well be encouraged by the teacher. It is an unfortunate fact that in reference to a course in botany the notion largely prevails that it consists of lesson-work with a text-book like a course in history or algebra. It is often more dreaded than the latter because of the supposed necessity of learning a long list of difficult technical terms. Few teachers would be willing to give up the use of the text-book entirely and it is not at all necessary that they should. But every teacher can now choose a modern book of botany from the fairly long list that is offered by American publishers. These are not mainly terminology nor written with the chief aim of enabling the pupil, after having gone through a sufficient number of chapters, to "analyze" flowers. Many of them unfortunately provide no means of identifying the native plants as a part of a school course, but teachers are not left without choice of a good book after such ultra ones are thrown out of the list.

The text of an elementary book on botany should contain the important facts and principles of the science, and give a brief but comprehensive idea of the plant kingdom, in simple and plain language. An intimation and partial elucidation of means and methods employed to test or to verify the principles and inferences should be evident in the text. But this of itself is not sufficient for pedagogical purposes; there should be besides practical work provided, regular in time, ample in amount, that may train in the exercise of observation, experimentation and judgment.

I have for years devoted one-half the time of the botanical courses, both elementary and advanced, to such real work carried on partly in the laboratory, partly in the field. Besides courses here referred to others devoted wholly to laboratory, experimental or observational work are provided; but it is not my purpose to discuss these now. Neither is it necessary to give here a detailed outline of the practical work that should constitute a substantial portion of the elementary work for beginners. Those who wish to use such a simple yet ample course in the public schools can consult the "Practical Studies in Elementary Botany" published by Eldredge & Bro., Philadelphia, Pa.

But I desire to say in this connection that more real work on the native flora than is attempted even by able and enthusiastic teachers in Ohio schools would undoubtedly be advisable. I have outlined some competition work and submitted it to some of the schools looking to more interest in elementary practical work in this science. It has been urged that the project might be made more widely known to our Ohio schools with possible advantage, and therefore I have furnished, though with some misgivings, the following statement of this scheme.

Either of the following subjects may be selected: Mosses, Lichens, or Trees; the work to conform to the suggestions and directions given below. The Report of the work must be completed on or before May 15, 1901, and submitted to the Teacher of Botany, or person (or persons) designated by him, who—taking into account both the quality and quantity of the work—will forward, if worthy, the best report accompanied by the illustrative material, to the undersigned; whereupon the latter will, on or before May 31, send as a reward to the author of said report a copy of the OHIO NATURALIST VOL. 1.

Pupils now studying, or those who have formerly studied, botany are eligible to enter the competition. No award will be made unless at least two or three pupils undertake the work; it is hoped that every member of the class will compete.

It is desired that the pupils consult teachers, parents, and others, who may be able to advise as to the subject, kind and extent of the work, also as to the best arrangement and wording of the report, and the labelling and preparation of the accompanying illustrative material.

The report is to contain a detailed account of the work actually done by the pupil and in no case to contain anything not his own.

The names of those entering the competition must be sent to the undersigned on or before March 30th. The suggestions, directions and explanation of the three subjects proposed are as follows:

BRYOLOGICAL.—All the kinds of Mosses in the region should be collected and put under slight pressure till dry; then a small portion should be glued directly upon a piece of card-board and a larger amount placed in a paper pocket and attached to the same piece; the notes and drawings can also be attached to the same card-board which for each species should be $8\frac{1}{4} \times 11\frac{1}{2}$ inches. Most of the kinds (species) can be found in fruit; the latter is a capsule (little pod) on a slender stem called the seta. Specimens without fruit are not very satisfactory.

Tell in each case on what the specimen grows, as the ground, tree trunk, old log, rock, boulder, etc.; add other notes relating to its situation (habitat), abundance, appearance, general character (habit), etc.

Draw an enlarged figure at least of the capsule (fruit) of some or all of the species (kinds) collected. In the early stage there is usually a cap (called calyptra) on the capsule. When the capsule is ripe it opens by a lid (called the operculum) for the escape of the spores. Notice the teeth (called collectively the peristome) surrounding the mouth of the capsule—evident when the operculum falls off.

A good pocket lens must be used for this work. The drawings must be clear; after completed with a sharp lead pencil it would be well to retrace with a fine pen and india (or drawing) ink. Excessive shading of the figures is objectionable.

If a book is desired, a suitable one for beginners is Grout's "Mosses with a Hand-lens," price \$1.10; orders sent to the author or to the writer of this article will be promptly attended to. But for the purposes of this competition the botanical names of mosses are not *required*; it will be of course more interesting if an attempt at the identification of the species is in all cases made.

LICHENOLOGICAL.—All the kinds of Lichens in the region should be collected. The little disks, or saucer-like bodies, on the plants are the fruit (called the apothecium); the apothecia are more distinct and striking in appearance, as is the whole plant also, when moist; therefore the best time

to collect lichens is after a prolonged rain, or when the air is moist; when dry they are usually brittle and cannot be satisfactorily handled.

Do not save specimens that have no fruit, except in case of rare species. Only enough pressure on the specimens (placed between blotters or soft papers) should be brought into requisition as is necessary to keep them from curving or crumpling while drying. Then glue a specimen to a card-board, $8\frac{1}{4} \times 11\frac{1}{2}$ inches, and also attach a paper pocket containing ample material, and the drawings (if any are attempted), also the notes, to the same piece of card-board. Use a separate card-board for each kind (species).

Tell the substratum on which the specimen was found—as boulders, limestone, sandstone, log or stump, fence-rail, tree or plant, soil, etc. Give additional notes as to appearance, size, abundance, habitat, habit, etc. Those growing on rocks can not generally be removed—a thin piece of rock must be chipped off to secure them.

A detailed description should be written of each kind (species); drawings perhaps might be undertaken; the different species should be compared and contrasted. Use a good pocket lens. There is no text-book on Lichens that is usable by beginners.

DENDROLOGICAL.—The Trees may be studied from one of several points of view. If a camera be used, selected trees should be studied and illustrated; the bark compared in case of different species, likewise in case of one and the same species when the individuals are of different ages and sizes or grow in different situations or exposures; also modes of branching compared and shapes contrasted. Very full notes should be taken, and when written up in the report reference should be made constantly to the numbered illustrations. Few or many kinds of trees, as preferred, may be taken if this phase of the subject is selected.

Instead of the above one may study and identify all the kinds (species) of trees in the region. Full descriptions should be written out, and similarities and contrasts of different species noted. Give uses of the kinds of woods only when such use is made in the region or the near town or city. Collect twigs and fasten them to card-boards ($8\frac{1}{4} \times 11\frac{1}{2}$ inches). Attach a specimen of the fruit also when it can be found under the tree. A pamphlet (price 10 cents) with a Key to the Ohio Forest Trees by means of which the names can be determined, may be obtained from the writer.

A third method of carrying out the work on trees would be to give an account of the forest area in the region—either taking a square or rectangular tract of a mile or more in extent; or selecting if possible a natural area, as a river or creek valley, or other obviously bounded tract of ample dimensions. Draw a map of the selected region and locate thereon the forests and groups of trees. Describe them, indicating the prominent kinds of trees, the less abundant species, and the very rare ones. Tell approximately the size of the largest, the commonest size, etc. Note uses made of some of the kinds in the region or at a near manufactory. Record other observations.

MINOR PLANT NOTES, NO. 2.

W. A. KELLERMAN.

TARAXACUM ERYTHROSPERMUM.—The Red-seeded Dandelion, now known to be common in our State, is a late bloomer. An abundance of flowers may be seen way after the severe frosts of autumn set in. Mr. Fred. J. Tyler collected specimens in bloom at Perry, Lake County, December 17. He reports “great fields” of it at that place, whereas the common Dandelion (*Taraxacum taraxacum*) was conspicuous by its absence. Prof. Beardslee of Cleveland, reports the Red-seeded form as the one of common occurrence in Cuyahoga County. I have noted the Red-seeded form in bloom near the city of Columbus December 23, though the month has been a cold one, the thermometer registering once 10° F. The Common Dandelion (*Taraxacum taraxacum*) does not seem to bloom so late in the season—at least it is in bloom much less abundantly here. Contributions of phenological observations on interesting plants of our flora by readers of THE NATURALIST are in this incidental way earnestly solicited.

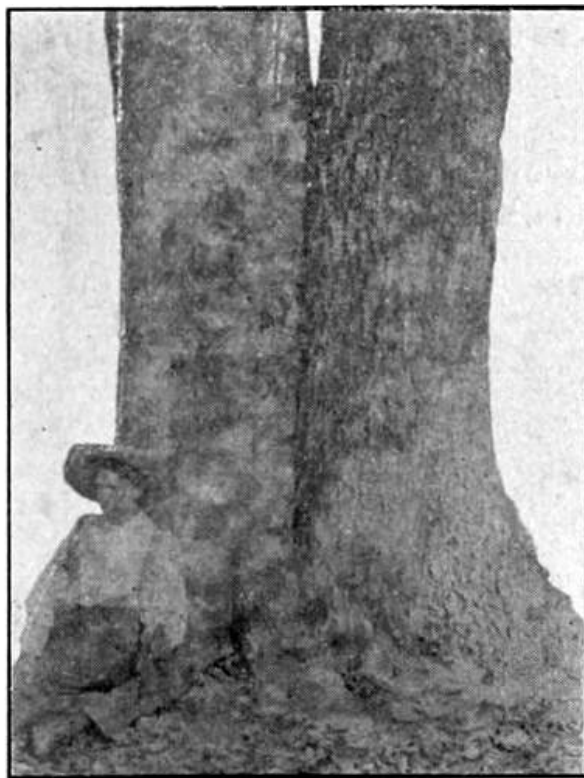
GROVE OF LARGE BEECHES.—There are now remaining in Ohio very few large groves of beeches. Of groves of very large beeches the same may be said. At Arion, in Scioto County, in the narrow valley of Brush Creek, are a large number of magnificent specimens of this very attractive American tree. The trunks are straight as is always the case for this species, smooth, and many of them are ten to twelve feet in circumference. One specimen measured twelve feet four inches, three feet from the ground. The grove is now used for picnic and camping purposes, and it is sincerely hoped that these splendid trees may be sacredly preserved for an indefinite time.

HABITAT OF RHAMNUS CAROLINIANA.—The manuals give the habitat of this species “in swamps and on low grounds” (Britton), “swamps and river

banks” (Gray), “river banks” (Wood), and “fertile soil” (Chapman). It has been previously reported that this species was found in Ohio last November. Several specimens were found near the Ohio river in Adams county at the mouth of Brush creek, and a few were seen in Brown county. In a little ravine on Cedar creek, a tributary of Brush creek, in Adams county, fourteen miles north of the Ohio river, an enormous number of plants were growing. Some were nine to eleven inches in circumference at the base and fourteen feet high. None occurred in “swamps,” though many were in “low grounds” where the soil seemed to be fairly fertile. The majority were on rocky hillsides or quite on the top of very high ground. The annexed cut shows a specimen near the top of a rocky bluff or hill perhaps one hundred and fifty feet above the valley at Cedar Mills, Ohio. This southern Buckthorn still retained its shining leaves though my visit was late in November when nearly all the other trees except the oaks were bare. This, with the great quantities of black fruits, presented a charming spectacle. The plant is also reported in Stanley Coulter’s catalogue of Indiana plants, discovered in the southern counties by Mr. W. T. Blatchley, “growing on rocky hillsides.”



Rhamnus Caroliniana on a rocky hill.



Twin Beech and Red Oak.

TWIN TREES; TWO SPECIES.—Sometimes two trees attempt to occupy the same space at the same time. The cut above shows a red oak and a beech in close juxtaposition, neither having been able to crowd the other out, and the two are united for a short distance from the ground. This would hardly be called a natural graft perhaps, though the two are intimately united. The trees are vigorous typical specimens of the two species, growing near Brush creek, at Arion, in Scioto county, Ohio. Several other examples in the same region were noticed. Sometimes the two trees are the same species, but usually of different species, the union of tissue in all cases equally evident.

MEETING OF THE BIOLOGICAL CLUB.

A meeting of the Biological Club was held in Zoological Lecture-room January 7, 1901. Professor Osborn presided; twenty-three members present.

Mr. Griggs, secretary of the board of editors gave a financial report and recommended that the name of the publication be changed to THE OHIO NATURALIST. By the unanimous vote of the members present, the recommendation was adopted.

Professor Hambleton read a paper on his Explorations in southern Chile and Patagonia. He said:

The expedition was sent out by the Chilean government and had for its object the exploration of the coast from about 40° south latitude to 48° in search of a river that might come from beyond the mountains. This was done and a river called by the party *Baker River* was discovered emptying into Baker channel at about 47° 20' south latitude. This river was followed in its course to the water-shed and was found to be the outlet of Lake Cochrane, though its principal source was not discovered.

A study of the flora of the region traversed by the party brought to light the following facts: That the Peninsula of Taitao marks the division line between the rich and varied flora of the Llanquihue region and the monotonous flora of the Magellan Strait region.

The difference between these two floras is really remarkable. In the Llanquihue region no one species nor even genus can be said to predominate. Character is given to the vegetation by the *Chusquea quila* a sort of climbing bamboo. *Fuchsia macrostemma* and a large variety of stately forest trees, all struggling together for the mastery and making a forest as impenetrable as the Selvas of the Amazon.

The Magellanic region is characterized by the predominance of several species of *Nothofagus*. Immense forests may be found composed exclusively of a single species, for example, the *N. pumila*, and in all this region, extending from the Peninsula of Taitoa to Cape Horn, a distance 10° of latitude, no other kind of tree acquires any considerable size.

Professor Mills gave a report of the Baltimore meeting of Anthropologists, and Professor Osborn gave a short account of a visit with Dr. P. R. Uhler, and called attention to some of the Zoological papers read before the meeting of the Society of Naturalists.

J. K. Knox, Miss Mary Dresbach, Miss Mary C. Crawford, E. D. Coberly, C. C. Poindexter, A. F. Conradi and Miss Clara Tangeman were elected to membership.

The Society then adjourned.

JAS. S. HINE, Secretary.

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